

22AI503 – MACHINE LEARNING

UNIT 1 – INTRODUCTION

TOPIC 3: Relationship between attributes using Covariance and Correlations

3.1 Covariance

Covariance and **correlation** are statistical measures used in machine learning to understand relationships between variables (features). They are important for feature analysis, dimensionality reduction, and detecting redundancy in data.

Covariance measures how two variables change together.

For variables X and Y:

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Interpretation

- **Positive covariance (> 0):**
 - When X increases, Y tends to increase.
- **Negative covariance (< 0):**
 - When X increases, Y tends to decrease.
- **Zero covariance (≈ 0):**
 - No linear relationship.

Limitation

Covariance depends on the units of measurement.

For example:

- Height in centimeters vs meters changes the covariance value.
- Therefore, covariance is difficult to compare across datasets.

3.2 Correlation

Correlation is a standardized version of covariance.

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Where:

- σ_X = standard deviation of X
- σ_Y = standard deviation of Y

Range

$$-1 \leq \rho \leq 1$$

Example

- Study hours and exam scores → strong positive correlation.
- Product price and demand → often negative correlation.

3.3 Covariance vs Correlation

Aspect	Covariance	Correlation
Measures	Joint variation	Strength and direction of relationship
Range	Unbounded	-1 to +1
Unit dependent	Yes	No
Easy to interpret	No	Yes
Comparison across variables	Difficult	Easy

3.4 Relationship between Attributes using Covariance and Correlation

In machine learning and statistics, **attributes (features)** are often analyzed to determine whether they are related to each other. Covariance and correlation are two common measures used for this purpose.

1. Using Covariance to Measure Relationships

Covariance tells us whether two attributes move in the same direction or opposite directions.

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Interpretation

- **Cov(X,Y) > 0** → Positive relationship
 - Both attributes increase or decrease together.
- **Cov(X,Y) < 0** → Negative relationship
 - One attribute increases while the other decreases.
- **Cov(X,Y) ≈ 0** → No linear relationship.

Example

Study Hours (X)	Exam Score (Y)
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2	50
4	60
6	75
8	85

As study hours increase, exam scores increase. Therefore, covariance is positive.

2. Using Correlation to Measure Relationships

Correlation standardizes covariance and provides a value between **-1** and **+1**.

$$\rho_{XY} = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

Interpretation

Correlation (r)	Relationship
+1	Perfect positive relationship
+0.7 to +0.9	Strong positive relationship
+0.3 to +0.7	Moderate positive relationship
0	No linear relationship
-0.3 to -0.7	Moderate negative relationship
-0.7 to -0.9	Strong negative relationship
-1	Perfect negative relationship

Visual Example of Correlation

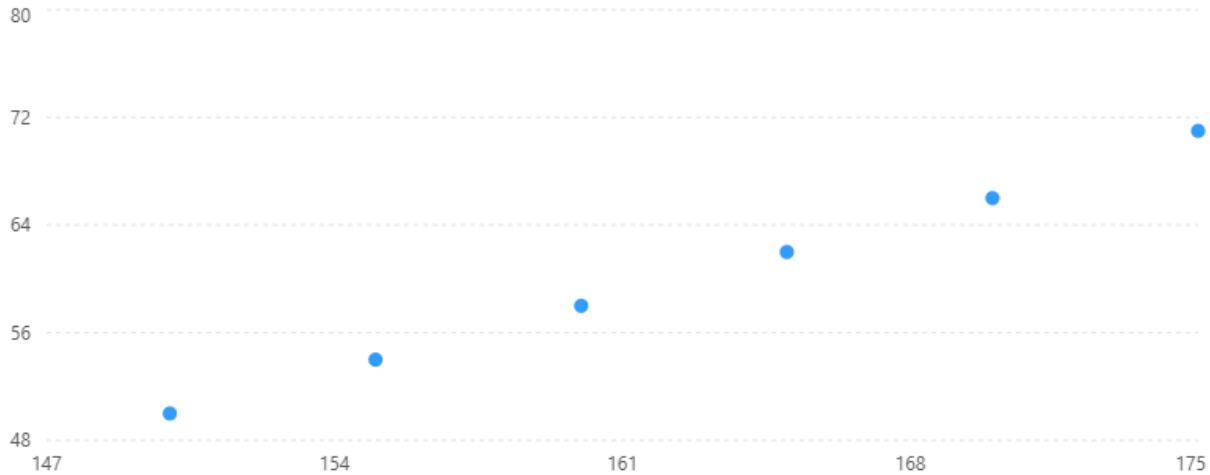
A scatter plot with a strong positive correlation might look like this:

Height vs Weight

Example of a strong positive correlation between two attributes.

Height vs Weight

Example of a strong positive correlation between two attributes.



As height increases, weight also increases, indicating a strong positive correlation.

3. Relationship Analysis in Machine Learning

Highly Positively Correlated Attributes

Example:

- Height and Weight
- Age and Work Experience

These attributes tend to increase together.

Highly Negatively Correlated Attributes

Example:

- Product Price and Demand
- Distance from Wi-Fi Router and Signal Strength

One attribute increases while the other decreases.

Uncorrelated Attributes

Example:

- Shoe Size and Exam Marks

No meaningful linear relationship exists.

4. Correlation Matrix

When a dataset contains many attributes, we calculate pairwise correlations.

Feature	Age	Income	Experience
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Age	1.00	0.60	0.90
Income	0.60	1.00	0.70
Experience	0.90	0.70	1.00

Here:

- Age and Experience have a very strong relationship (0.90).
- Income is moderately related to both.

3.5 Solved Problems

Problem 1: Calculate Covariance

Given:

X	Y
1	2
2	4
3	6
4	8
5	10

Step 1: Calculate Means

$$\bar{X} = \frac{1 + 2 + 3 + 4 + 5}{5} = 3$$

$$\bar{Y} = \frac{2 + 4 + 6 + 8 + 10}{5} = 6$$

Step 2: Create Table



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X	Y	$X - \bar{X}$	$Y - \bar{Y}$	Product
1	2	-2	-4	8
2	4	-1	-2	2
3	6	0	0	0
4	8	1	2	2
5	10	2	4	8

$$\sum (X - \bar{X})(Y - \bar{Y}) = 20$$

Step 3: Covariance

$$\text{Cov}(X, Y) = \frac{20}{5-1} = 5$$

Answer

Covariance = 5

Since it is positive, X and Y move together.

Problem 2: Calculate Correlation Coefficient

Data:

X	Y
1	2
2	4
3	6
4	8
5	10

Step 1: Covariance

From Problem 1:

$$\text{Cov}(X, Y) = 5$$

Step 2: Standard Deviations



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For X:

$$\sigma_X = \sqrt{\frac{10}{4}} = \sqrt{2.5} = 1.581$$

For Y:

$$\sigma_Y = \sqrt{\frac{40}{4}} = \sqrt{10} = 3.162$$

Step 3: Correlation

$$r = \frac{5}{(1.581)(3.162)} \approx 1.0$$

Answer

Correlation = +1

This indicates a perfect positive linear relationship

Problem 3: Negative Correlation

Data:

X	Y
1	10
2	8
3	6
4	4
5	2

Observation

As X increases, Y decreases.

Result

$$r = -1$$

Answer

Perfect negative correlation.

Problem 4: Find Covariance

Data:



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X	Y
2	1
4	3
6	5

Step 1: Means

$$\bar{X} = 4$$

$$\bar{Y} = 3$$

Step 2: Deviations

x	y	$X - \bar{X}$	$Y - \bar{Y}$	Product
2	1	-2	-2	4
4	3	0	0	0
6	5	2	2	4

$$\sum Product = 8$$

Step 3: Covariance

$$Cov(X, Y) = \frac{8}{3 - 1}$$

$$Cov(X, Y) = 4$$

Answer

Covariance = 4

Positive relationship exists.

Problem 5: Interpret Correlation Values

Suppose the following correlations are obtained:

Features	Correlation
Height vs Weight	0.88



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Price vs Demand	-0.75
Shoe Size vs IQ	0.02

Interpretation

- 0.88 → Strong positive relationship.
- -0.75 → Strong negative relationship.
- 0.02 → Virtually no linear relationship.

Problem 6: Correlation Matrix Interpretation

Feature	Age	Salary	Experience
Age	1.00	0.60	0.92
Salary	0.60	1.00	0.70
Experience	0.92	0.70	1.00

Questions

1. Which pair has the strongest relationship?
2. Which pair may cause multicollinearity?

Solution

- Strongest correlation = Age and Experience (0.92).
- Since $0.92 > 0.8$, Age and Experience may create multicollinearity in regression models.

DISCUSSION QUESTIONS:

1. What is the difference between covariance and correlation, and why is correlation often preferred when comparing relationships between attributes?
2. Can two variables have a covariance of zero but still be related? Explain with an example.
3. Why does a high correlation between two features sometimes create problems in machine learning models such as linear regression?
4. Correlation does not imply causation. What does this statement mean, and can you provide a real-world example where two variables are highly correlated but one does not cause the other?
5. How can covariance and correlation help in feature selection and dimensionality reduction techniques such as PCA in machine learning?
6. **Exercise 1: Calculate Covariance**



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Given the following data:

X	Y
2	1
4	3
6	5
8	7

Task: Calculate the covariance between X and Y and interpret the result.

7. Exercise 2: Calculate Correlation Coefficient

Given:

X	Y
1	3
2	5
3	7
4	9
5	11

Task: Find:

- A. Covariance of X and Y
- B. Standard deviations of X and Y
- C. Correlation coefficient r
- D. Interpret whether the relationship is weak, moderate, or strong.

8. Exercise 3: Negative Correlation

Given:

X	Y
1	12
2	10
3	8
4	6
5	4

Task:

- Determine whether the covariance is positive or negative.
- Estimate whether the correlation is close to +1, 0, or -1.
- Explain your reasoning.

9. Exercise 4: Correlation Matrix Analysis

Consider the following correlation matrix:

Feature	Age	Income	Experience
Age	1.00	0.55	0.91
Income	0.55	1.00	0.68
Experience	0.91	0.68	1.00

Questions:

- Which pair of attributes has the strongest relationship?
- Which pair shows a moderate relationship?
- Which features might cause multicollinearity in a regression model?
- Why?

10. Exercise 5: Real-World Interpretation

A company collected the following information:



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Variable Pair	Correlation
Advertising Spend & Sales	0.82
Product Price & Demand	-0.76
Employee Age & Productivity	0.05

Questions:

- Which pair has the strongest positive relationship?
- Which pair has a strong negative relationship?
- Which pair shows almost no linear relationship?
- How could this information help business decision-making?